

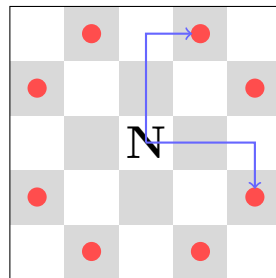
The Knight's Tour

1 The knight and its graph

A knight on square (x, y) may jump to any of the eight squares

$$(x \pm 1, y \pm 2), \quad (x \pm 2, y \pm 1) \quad (1)$$

that lie on the board. Abstracting the chessboard away, the $N \times N$ *knight graph* has one vertex per square and one edge per legal jump; everything below is graph theory in chess clothing.



The eight knight moves; two are drawn as their L shapes.

2 Open and closed tours

A *tour* visits every square of the board exactly once.

- An **open tour** may end anywhere: a *Hamiltonian path* of the knight graph.
- A **closed tour** (or re-entrant tour) ends a single knight's move away from its starting square, so the knight could go round forever: a *Hamiltonian cycle*.

Existence is completely settled on square boards:

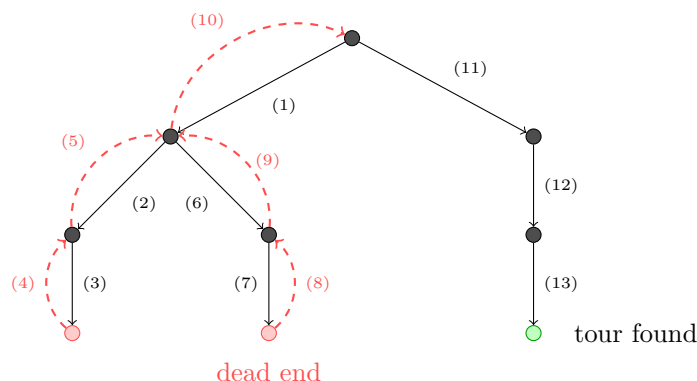
- Open tours exist exactly for $N = 1$ and $N \geq 5$. There is no tour on 2×2 or 3×3 (on the latter, the center square has no knight move at all), and — less obviously — none on 4×4 either.
- Closed tours exist exactly for even $N \geq 6$.

The parity half of the closed-tour statement has a one-line proof worth savoring. Every knight move changes square color, so any walk alternates black and white. A closed tour is a cycle through all N^2 squares using each exactly once; alternation around a cycle forces equally many squares of each color, hence N^2 even, hence N even. For odd N the board has one color in excess and no closed tour can exist — an impossibility established without examining a single candidate tour, just from an invariant every move preserves.

3 Algorithms

3.1 Backtracking

The direct search extends a partial path square by square: from the current square, try each unvisited neighbor in turn, recurse, and undo on failure; success is a path of length N^2 .



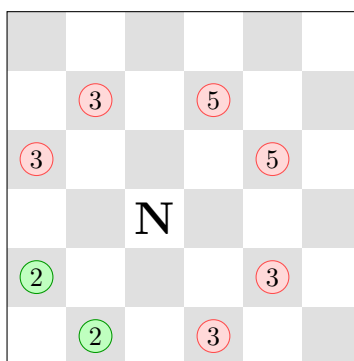
The search tree, edges numbered in traversal order: solid arrows descend, dashed red arrows undo after a dead end.

This always works in principle — it is exhaustive — but the knight graph is generous (up to eight choices per step, path length N^2), and from an unlucky start the tree above can grow astronomically many red leaves before its first green one. Unlike the queens problem, where each placement prunes entire columns and diagonals, a knight move constrains the future only weakly, so whole subtrees are explored in vain.

3.2 Warnsdorff's rule

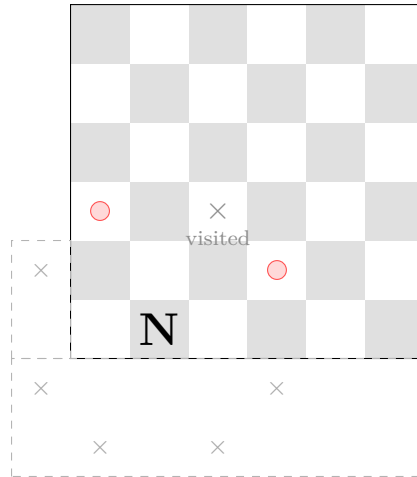
The classical remedy (1823) rests on one number. The *onward count* of a square: presume the knight stood there — to how many squares could it jump (on the board, not yet visited)? Warnsdorff's rule: *always jump to the candidate with the smallest onward count; ties are broken arbitrarily* (our program: first in its fixed move order).

Here is the rule at work on an empty 6×6 board; each candidate square is labeled with its onward count:



Warnsdorff picks a green square, whose onward count 2 is smallest.

Where does any label come from? Always the same arithmetic: a knight has *eight* potential targets; strike out those that fall off the board and those already visited; the survivors are the onward count. Here is that arithmetic for the bottom-left green 2, with the knight moved onto it and all eight of its targets drawn — including the five that land outside the board:



The green square's eight targets: five (gray $\times$) are off the board, one is already visited, and the two red squares survive — $8 - 5 - 1 = 2$, the label it carried in the previous figure. Every other label follows from the same subtraction.

The intuition is a rescue principle: the squares near the corner have the fewest ways in and out, so they are the ones in danger of becoming unreachable islands — visit them while it is still possible, and leave the well-connected center for later. With this ordering the very first descent of the search tree almost always runs straight to a green leaf: backtracking in name, linear in practice.

Two honest caveats, one of them visible above: the two green squares *tie* at count 2, and ties must be broken somehow — on larger boards a poor tie-breaking rule can strand the pure heuristic in a dead end. Keeping the backtracking machinery underneath restores certainty at no cost when the heuristic behaves. And greedy success is empirical, not guaranteed: the rule is a heuristic, not a theorem.

The closed-tour twist. The praise above is earned on *open* tours, where the rule asks nothing of the final square. A closed tour also demands that the last square rejoin the first, and a greedy descent satisfies that only by luck. Measurement was blunt about it: Warnsdorff-ordered *backtracking* solves 6×6 in seconds but churns indefinitely from 8×8 up — a prefix doomed not to close hides its failure astronomically deep in the tree, and backtracking near the leaves never repairs it. The effective remedy inverts the strategy: *restart* instead of persist — run the pure greedy descent with random tie-breaking, and when it fails to close, throw the path away and try afresh. Attempts are near linear and each closes with modest probability, so a tour arrives within milliseconds to seconds for $8 \leq n \leq 20$. The smallest board is the exception in the other direction: on 6×6 greedy descents essentially never close, while the ordered exhaustive search is quick — so the program uses each regime where it wins. The general lesson is worth keeping: a heuristic tuned for one goal (cover the board) does not automatically serve a stricter one (cover it *and* return home), and in heavy-tailed searches, restarting often beats persisting.

4 From mathematics to program

The program searches for *closed* tours. It represents the board as a visited matrix plus the trail of moves so far; the eight jump offsets are a constant list. A candidate is accepted when the path covers all N^2 squares *and* its last square is a knight's move from the first — the extra closure test is the only difference from an open-tour search. The result is an *option*: **Some tour** on first success — there is no point enumerating further — or **None**, which for odd N or $N < 6$ is not a failure of the search but a fact of mathematics. Sorting candidate moves by onward count

turns the naive search into Warnsdorff's; timing both against each other makes the difference measurable rather than anecdotal.